

Total No. of Printed Pages: 2

T.E - (Information Technology) (Sem-VI)(Revised Course 2019-2020)

EXAMINATION JUNE 2024

IT643- Computer Forensics and Cyber Security

[Time: 3:00 Hours]

[Max. Marks: 100]

Instructions: 1. Answer five questions. At least two from Part A, two from Part B and one from Part C.

2. Assume suitable data if necessary

Part A

- Q.1**
- a) Discuss the importance of Computer Forensics 4
 - b) What are the benefits of professional forensics methodology? How is forensics different from data recovery? 8
 - c) Elaborate on CFX- 2000 and use of forensics in Human resource/ Employment proceedings. 8
- Q.2**
- a) A cyber-crime has occurred in Pitfalls Pvt. Ltd. What should be the steps taken by the Computer forensic specialist to identify and retrieve possible evidence that may exist on the Computer system? 6
 - b) Compare the 3 types of evidence. Why is collecting evidence necessary? 6
 - c) Discuss any two Data backup obstacles. 4
 - d) Discuss Freezing the Scene as a method of gathering evidence. 4
- Q.3**
- a) Discuss some Computer evidence processing steps 8
 - b) Compare conventional tape backup and today's modern back-up systems. 4
 - c) Elaborate on the problems with Computer Forensic evidence. 3
 - d) Why the maintaining Chain of Custody important? 5

Part B

- Q.4**
- a) Discuss "Hacking methodology". 8
 - b) Elaborate on Low tech methods for Cracking Passwords 6

- c) Discuss the characteristics desired in the tools you select for Ethical hacking 4
- d) Identify any one password cracking tool 2
- Q.5** a) Explain in detail the 4 basic steps used by hackers to perform Social engineering. 8
- b) Write a short note on Password Vulnerabilities 4
- c) Why do hackers hack? Why has hacking become an easy task today? Discuss some methods of maintaining anonymity. 8
- Q.6** a) Elaborate on " Port scanners" 8
- b) Discuss the Denial of Service attacks. 6
- c) Outline some wireless network Vulnerabilities 6

Part C

Q.7 Write short notes on the following:

- a) Network analysers
- b) Social Engineering Countermeasures
- c) Shoulder surfing
- d) Script kiddies, Intermediate and Elite hackers

4X5=20

- Q.8** a) Identify the characteristics you need to keep in mind while selecting a Computer Forensic Specialist? 6
- b) Discuss BAIT and DIRT as Business Computer Forensics technology 8
- c) What is the use of Computer Forensics in Law enforcement? 6